Diplomacy Discourse & Conflict Forecasting

Early-warning insights from diplomatic language

By Berk Sankır

Research Question

Do weekly changes in diplomatic discourse from country i toward country j help predict protest or military/tension events in country j over the next week (and, conceptually, the next 1–4 weeks)?

The project is done in the context of Russia-Ukraine war, but methods can be applied to other events.

Objectives

• Investigate how changes in diplomatic discourse signal upcoming protests and military events

• Compare pre-war (2019–2021) and war (2022–2024) periods for event distributions

• Test whether peace, war, economy, support and condemn frames affect next-week events

• Evaluate predictive models trained on discourse features

Data & Methods

Datasets: Diplomatic texts (GDN), event records (GDELT) and context measures (Lowy Index) are integrated into a 2019–2024 dyad-week panel.

Features: Lexicon-based ratios capturing war, peace, economy, support, condemn, human rights and security frames; sentiment scores; lagged features; and text volume (num_docs).

Outcomes & Splits: Next-week protest and military count/binary outcomes; time-based train (≤2022), validation (2023) and test (2024) splits ensure no leakage.

H1: Pre-War vs War (Counts)

• Protest rates rise from 0.158 to 0.274 (χ^2 $p \approx 1.6 \times 10^{-5}$)

• Military/tension rates remain similar (0.326 vs 0.331)

• Count distributions shift significantly for both outcomes

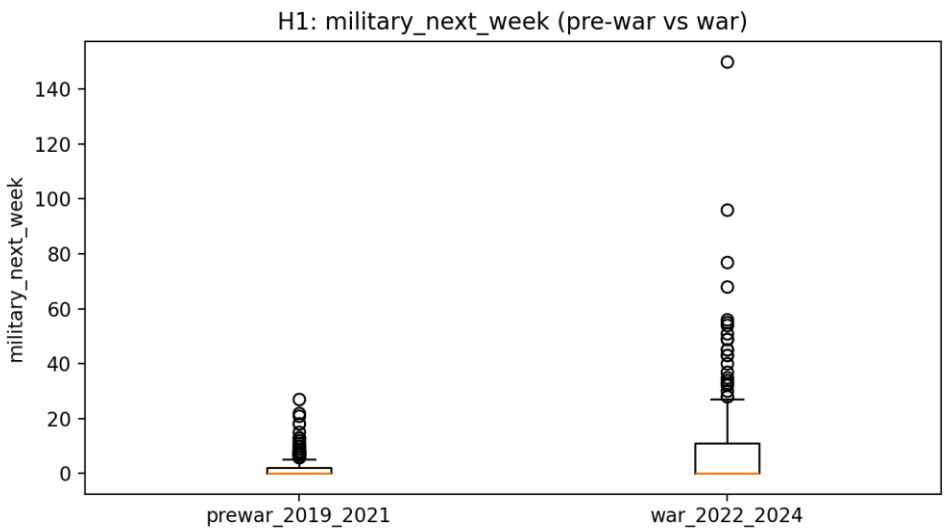


Figure: Military counts (Pre-War vs War)

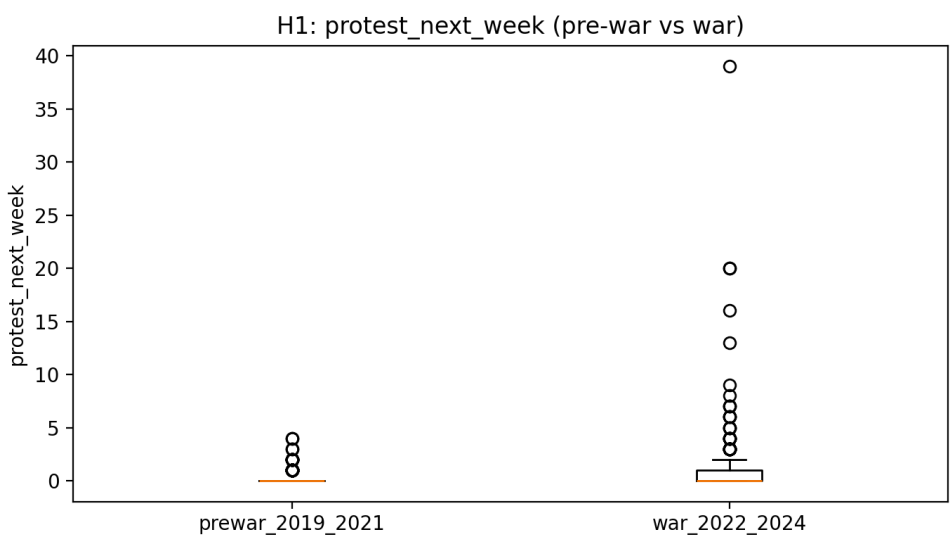
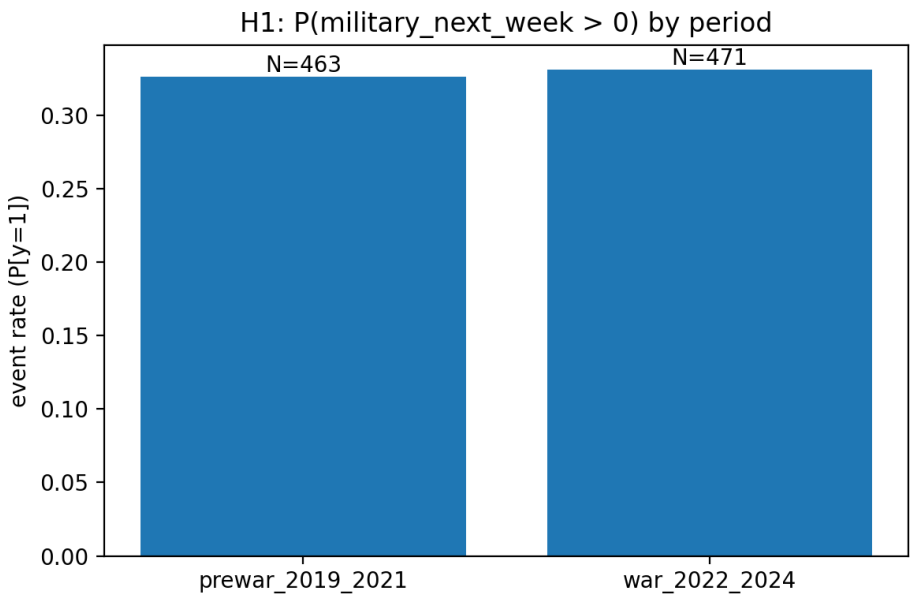


Figure: Protest counts (Pre-War vs War)

H1: Pre-War vs War (Event Rates)

- Probability of ≥1 military event more than doubles during war
- Probability of ≥1 protest event rises modestly



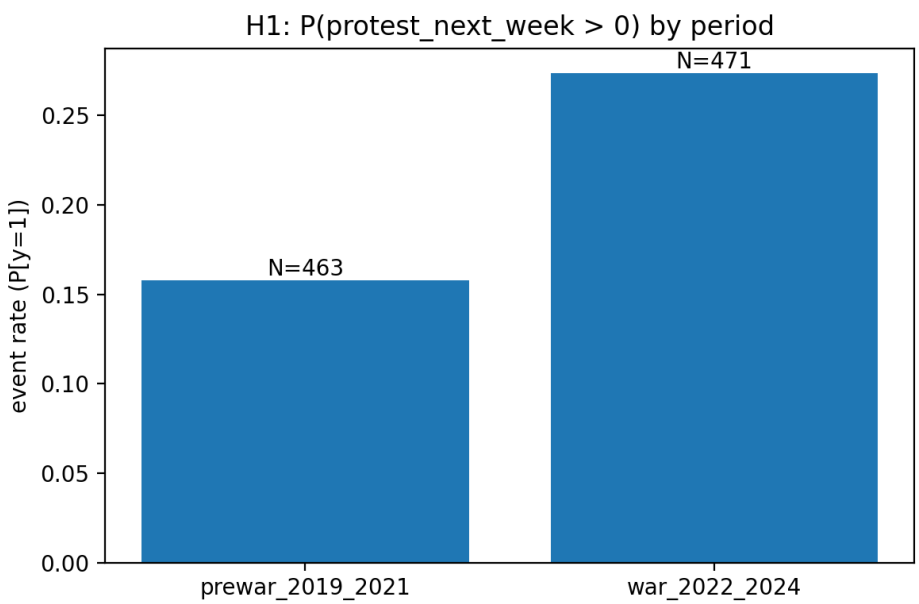


Figure: Share of dyad-weeks with at least one event (military and protest)

H2 & H3: Discourse Effects

H2 (Military): Peace and economy frames reduce next-week military events (IRR≈0.987). War/security/condemn frames show weak or no effect.

H3 (Protest): Support ratio and supportive tone strongly reduce protests (IRR≈0.68). Condemnations may slightly increase protest activity.

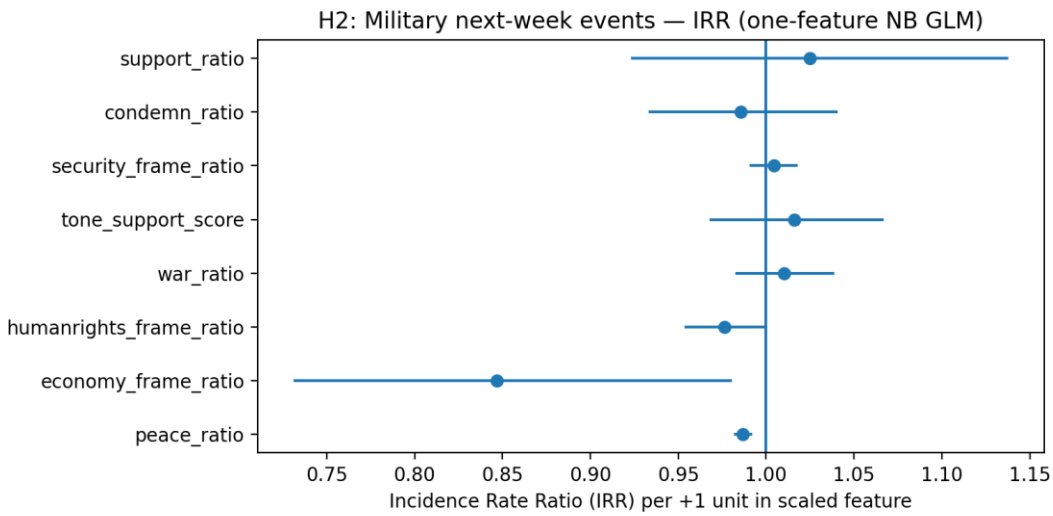


Figure: Military IRR plot

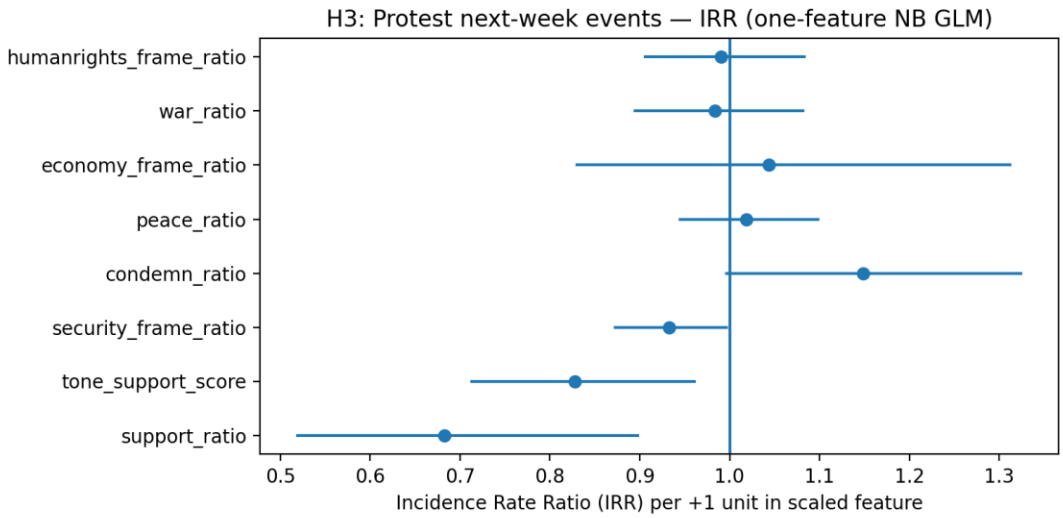


Figure: Protest IRR plot

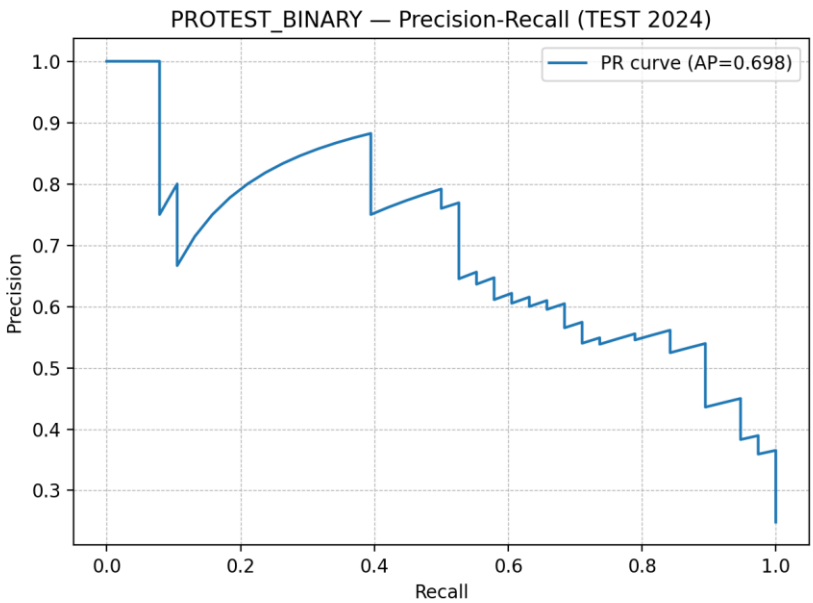
Model Performance

Binary classification (Test 2024):

• Protest model: ROC-AUC≈0.875; PR-AUC≈0.698; Precision≈0.538; Recall≈0.737; F1≈0.622

• Military model: ROC-AUC≈0.894; PR-AUC≈0.773; Precision≈0.598; Recall≈0.961; F1≈0.737

Count prediction: Negative Binomial GLM yields best MAE (0.572 protest; 4.03 military) and RMSE (0.934 protest; 5.57 military).



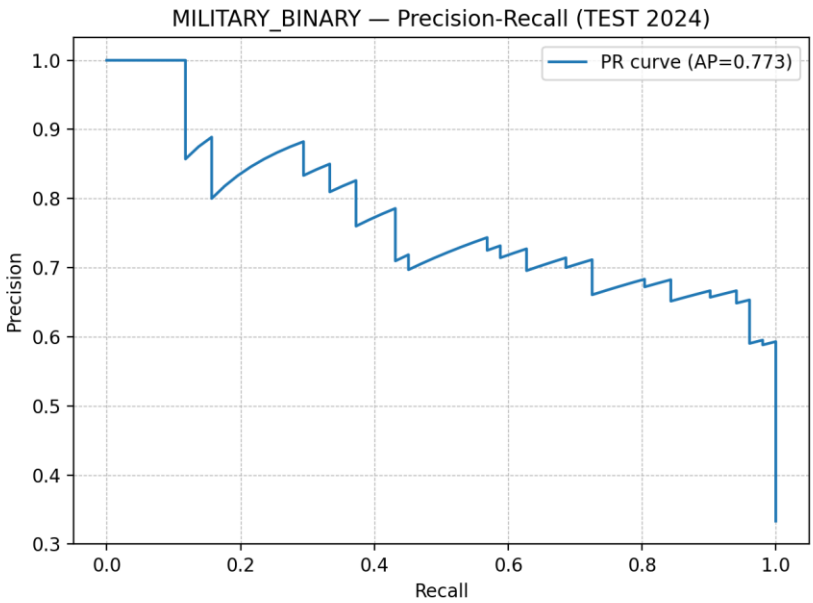


Figure: Precision–Recall curves for protest and military predictions

Limitations

The analysis is based on a limited number of dyads in a particular geopolitical environment. Thus, the results may not be representative for other geopolitical periods or regions. The lexicon features are designed to be simple. More sophisticated NLP features, such as context embeddings or supervised stance detectors, might lead to stronger features. The event labels are unbalanced and rely on GDELT data. Future studies may also investigate alternative event taxonomies, as well as forecast horizons such as t+2 to t+4, and causal modeling for additional dyads.

Conclusion

Weekly discourse signals related to diplomatic discourse have early warning value related to next week's events, but this varies depending on event type. For military/tension events, discourse features related to condemnation, security/war, and their lags have predictive value related to whether or not an event takes place. For protest events, predictability is possible, though dependent on text volume, or num_docs, suggesting that protest predictability is related to attention rather than content. Discourse features have early warning value, especially related to military/tension events, while also suggesting limitations related to protest predictability and count events.